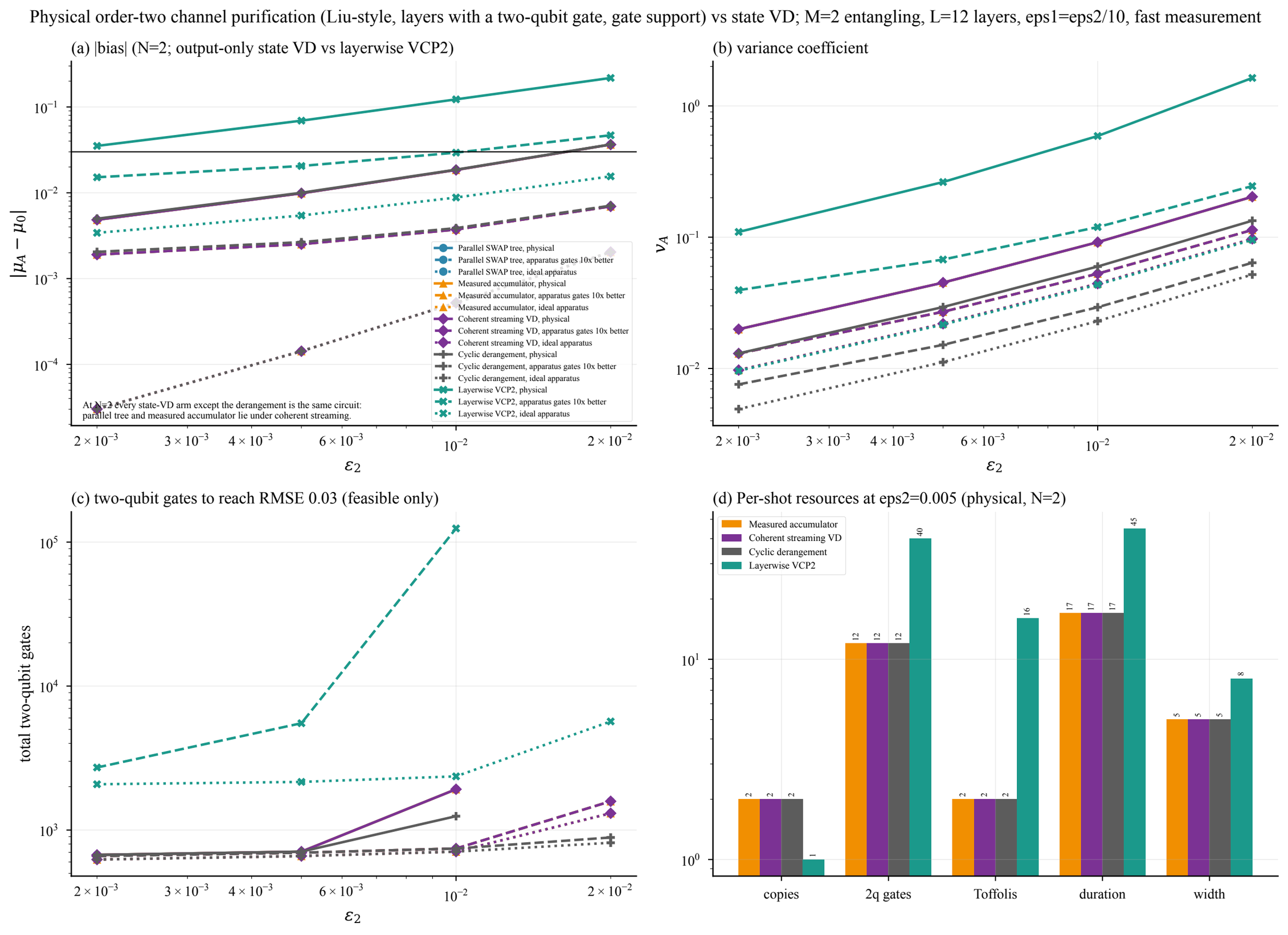

Physical order-two channel purification (Liu-style, layers with a two-qubit gate, gate support) vs state VD; M=2 entangling, L=12 layers, eps1=eps2/10, fast measurement
(a) |bias| (N=2; output-only state VD vs layerwise VCP2)
$|\mu_A - \mu_0|$
ε_2
At N=2 every state-VD arm except the derangement is the same circuit:
parallel tree and measured accumulator lie under coherent streaming.
Parallel SWAP tree, physical
Parallel SWAP tree, apparatus gates 10x better
Parallel SWAP tree, ideal apparatus
Measured accumulator, physical
Measured accumulator, apparatus gates 10x better
Measured accumulator, ideal apparatus
Coherent streaming VD, physical
Coherent streaming VD, apparatus gates 10x better
Coherent streaming VD, ideal apparatus
Cyclic derangement, physical
Cyclic derangement, apparatus gates 10x better
Cyclic derangement, ideal apparatus
Layerwise VCP2, physical
Layerwise VCP2, apparatus gates 10x better
Layerwise VCP2, ideal apparatus
(b) variance coefficient
v_A
ε_2
(c) two-qubit gates to reach RMSE 0.03 (feasible only)
total two-qubit gates
ε_2
(d) Per-shot resources at eps2=0.005 (physical, N=2)
Measured accumulator
Coherent streaming VD
Cyclic derangement
Layerwise VCP2
copies
2q gates
Toffolis
duration
width